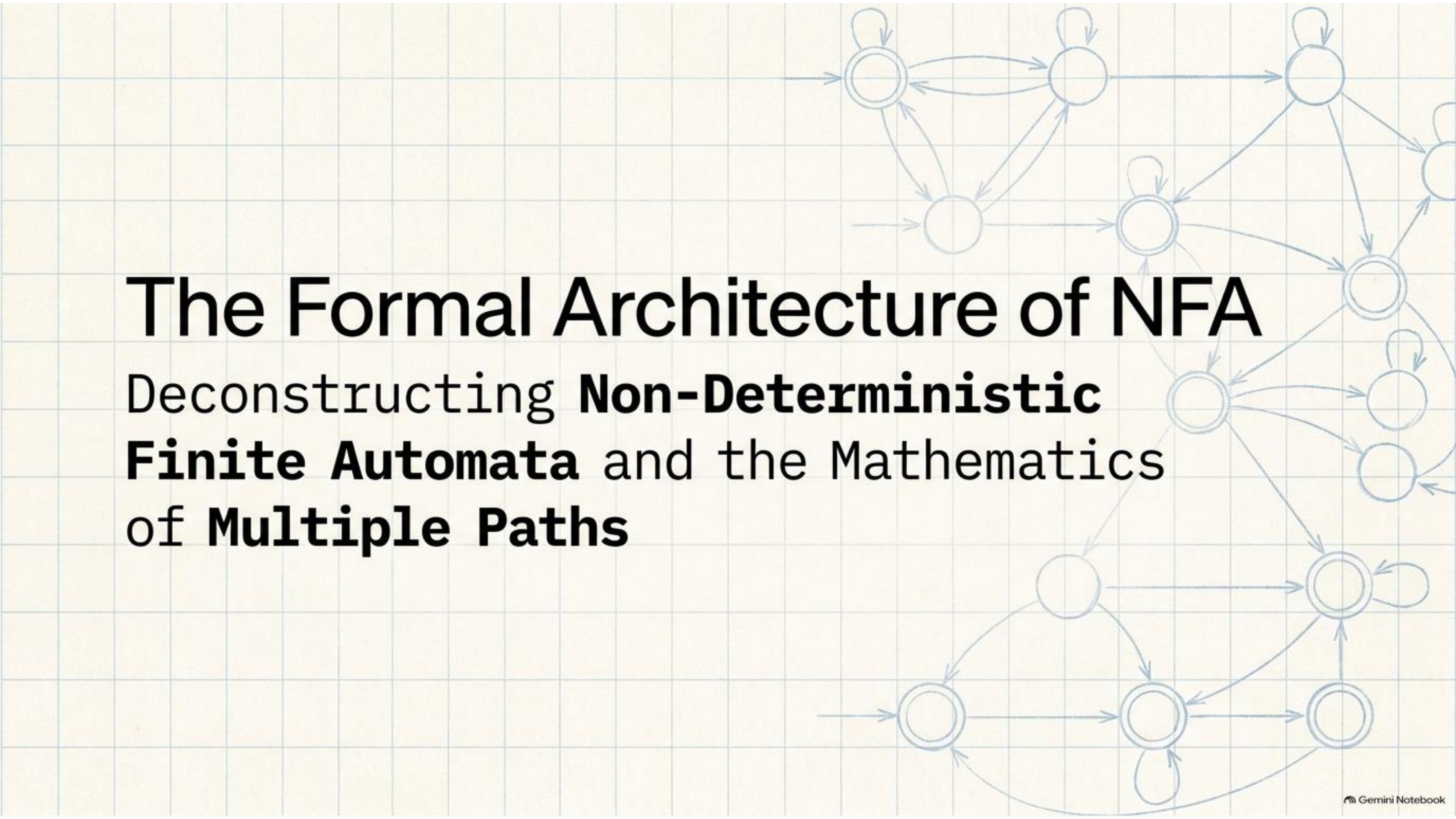
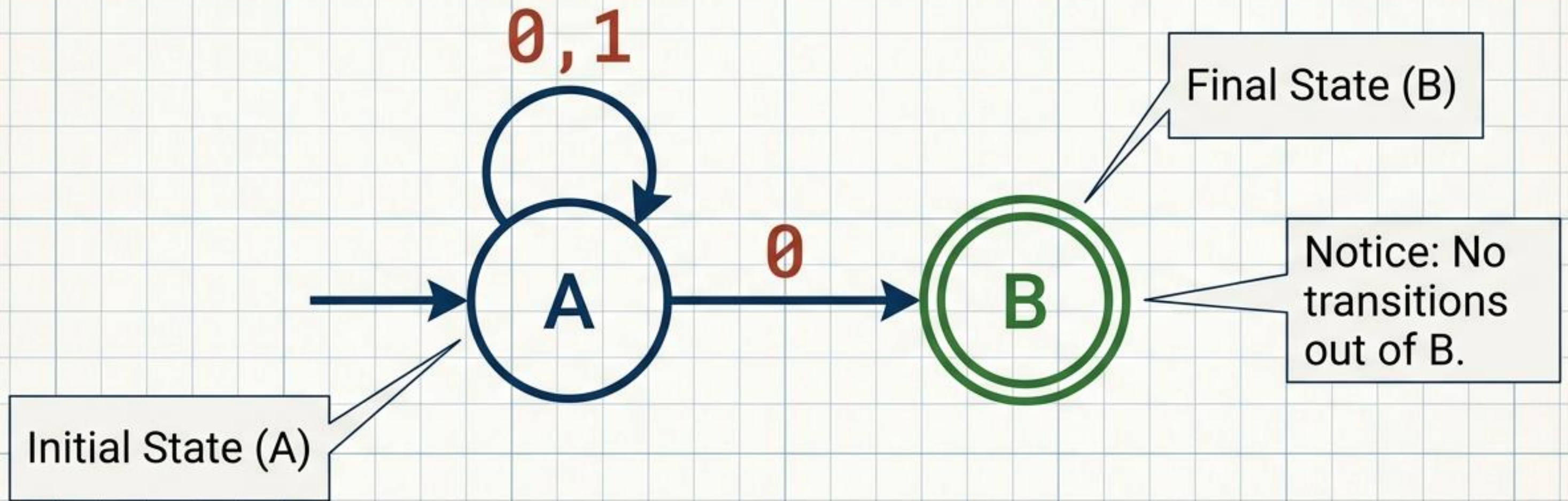




The Formal Architecture of NFA

Deconstructing **Non-Deterministic Finite Automata** and the Mathematics of **Multiple Paths**

A machine that accepts strings ending in "0".



The Functional Shift

	DFA (Deterministic)	NFA (Non-Deterministic)
Next State Logic	Strict 1-to-1 mapping. One input strictly leads to one next state.	Multiple paths. One input can simultaneously trigger multiple different next states.
Incomplete Transitions	Illegal. Every state must have defined transitions for every possible input.	Perfectly legal. States can completely lack transitions for certain inputs, leading nowhere ($\emptyset$).
Design Complexity	Rigid, complex, and requires careful mapping of all potential inputs.	Straightforward and flexible. You only build the paths you strictly need.

$$NFA = (Q, \Sigma, q_0, F, D)$$

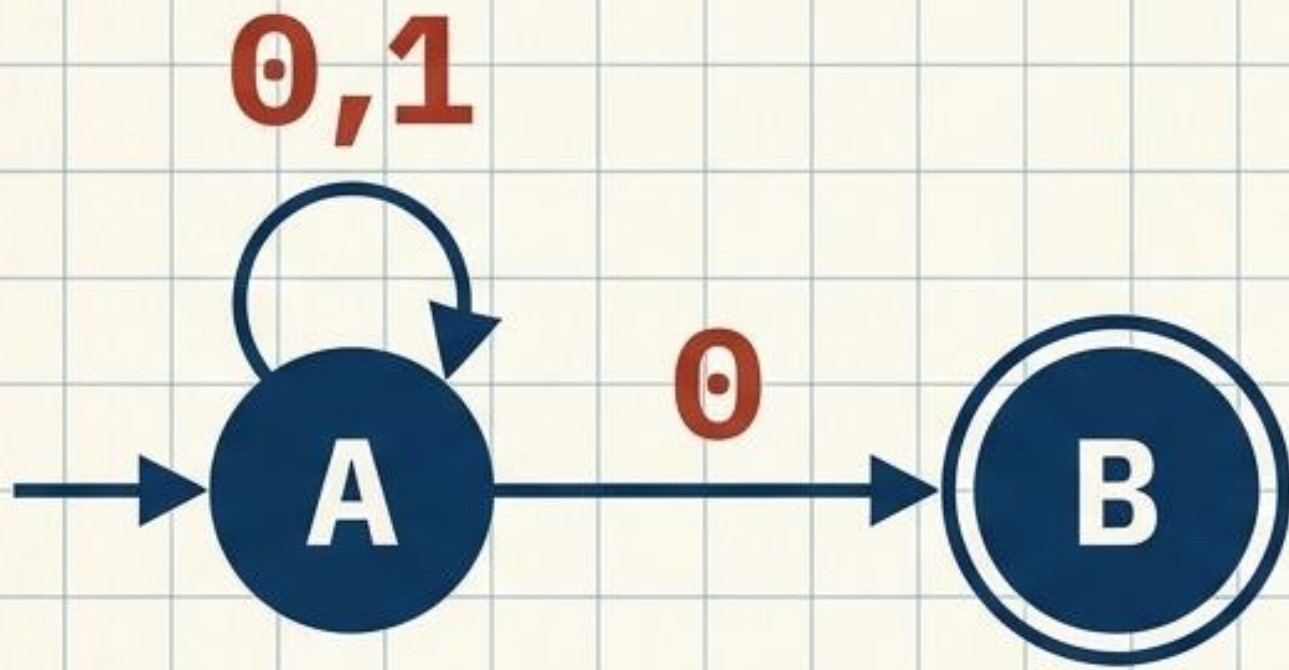
Set of all
states

The input
alphabet

The starting
state

Set of final
states

The
transition
function

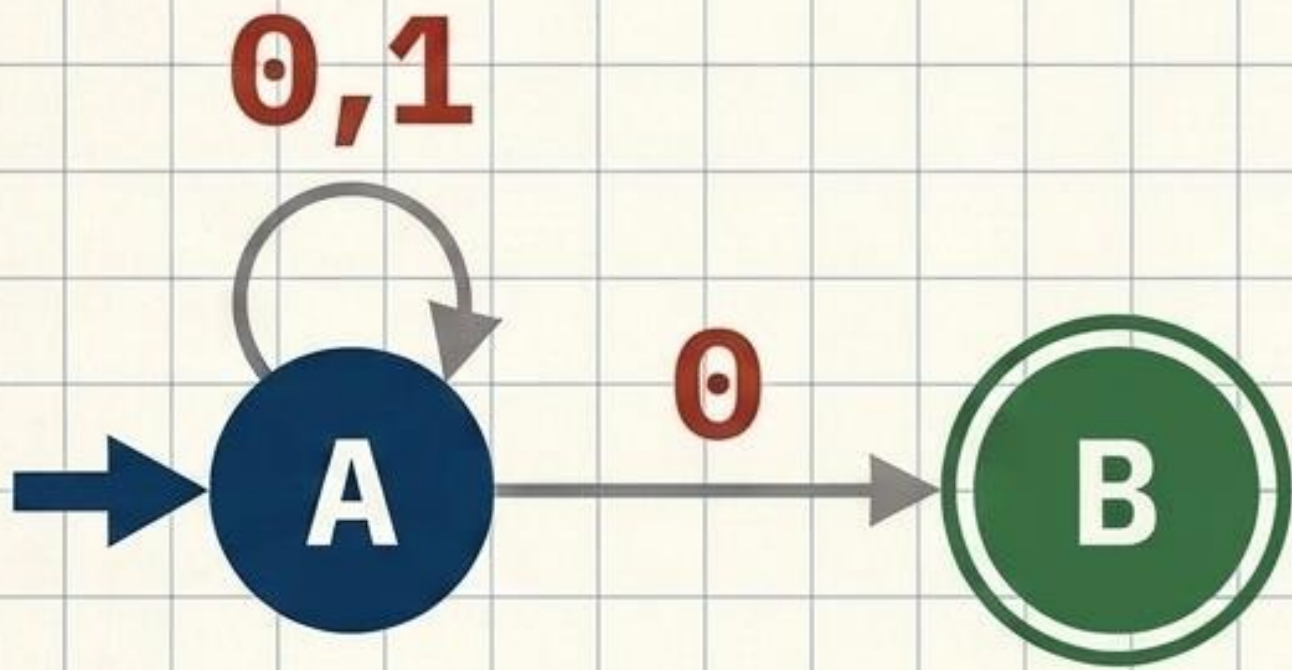


$$Q = \{A, B\}$$

The discrete set of all functional states within the machine.

$$\Sigma = \{0, 1\}$$

The definitive set of valid inputs the machine can process.



$$q_0 = A$$

The single initial state where sequence processing begins.

$$F = \{B\}$$

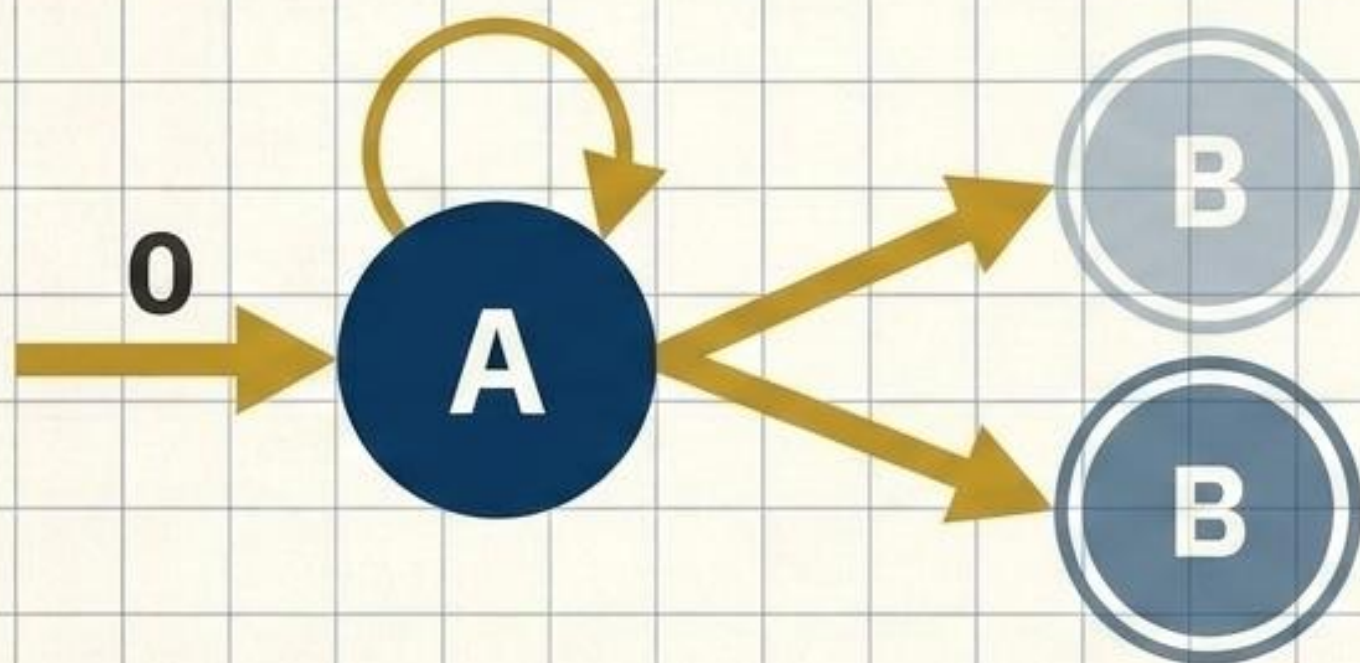
The set of final (or accepting) states.

(Q, Σ, q_0, F, D)

The Transition Function Anomaly

In a traditional DFA, D maps $Q \times \Sigma \rightarrow Q$. One state, one input, exactly one destination. But an NFA breaks this rule. Inputs can trigger multiple destinations, or lead nowhere at all.

So, where does an NFA transition function map to?



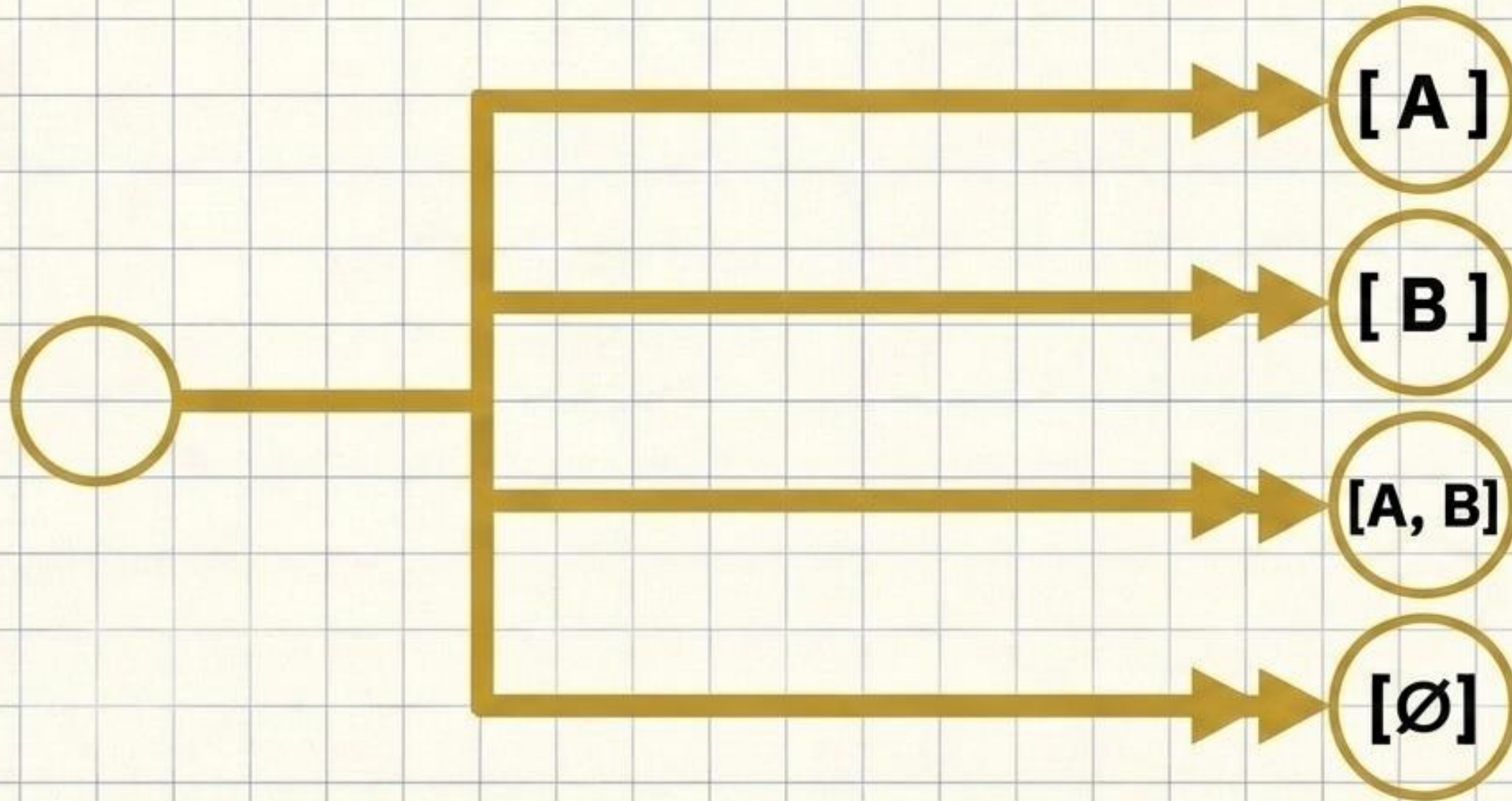
State A receives '0' →
Transitions to both A and B
simultaneously.



State B receives '0' or '1'
→ Transitions nowhere ($\emptyset$).

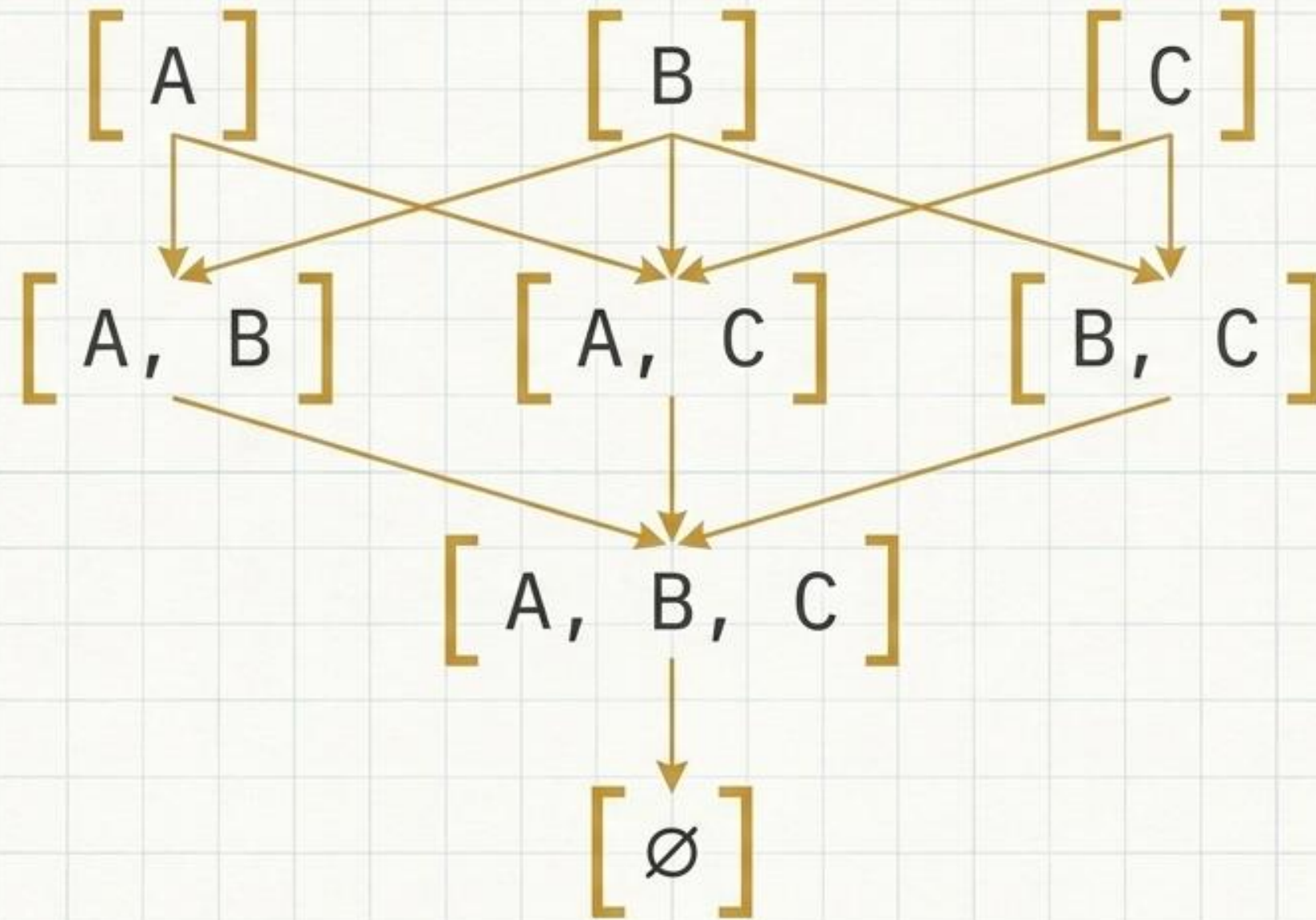
Given a state and an input, an NFA can jump to a single state, multiple states, or completely die out.

Mapping the 2-State Universe (A, B)



2 States = 4 Possibilities (2^2)

The 3-State Universe (A, B, C)



3 States = 8 Possibilities (2^3)

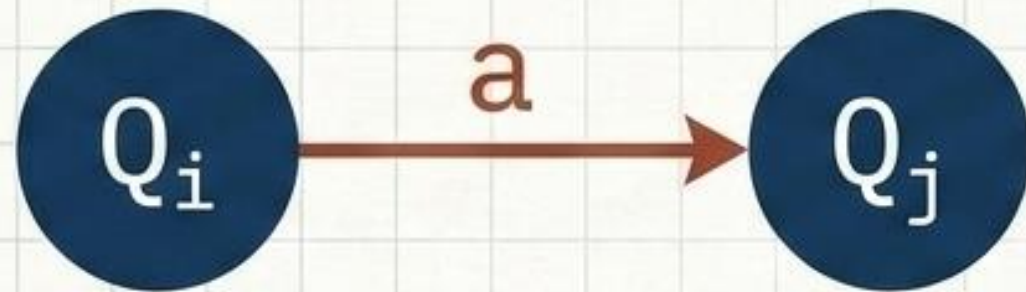
$$D: Q \times \Sigma \rightarrow 2^Q$$

The Power Set

Because an NFA can transition to any combination of available states—including all of them or none of them—the transition function maps to the Power Set of Q , denoted mathematically as 2^Q .

DFA: Deterministic Finite Automaton

$$Q \times \Sigma \rightarrow Q$$



NFA: Nondeterministic Finite Automaton

$$Q \times \Sigma \rightarrow 2^Q$$

